

SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

Sector	Residential
Building type	Multi-Unit
Building subtype	Duplex - Detached, conditioned basement (with main floor)
Vintage	1945-1985

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation and performance analysis.

BUILDING CHARACTERISTICS

Floor Area [m²]	349
Climate zone	7
HVAC SYSTEM	
Cooling system	Air-Source Heat Pump (Mini-Split)
Heating system	Electric Baseboards
Ventilation system	/
Typical COP cooling	2.50

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²/K]	0.55
Interior walls [W/m²/K]	0
Roof [W/m²/K]	0.3
Slabs [W/m²/K]	4.16
Infiltration (ELA @4Pa) [cm²]	361.0

GLAZING

Glazing [W/m²/K]	2.94
SHGC	0.6

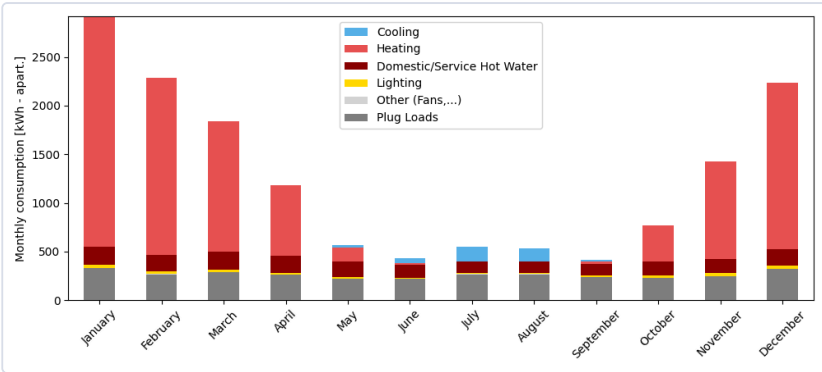
WINDOW-TO-WALL RATIO

Average [%]	5.9
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ANNUAL END USES — ELECTRICITY [kWh]

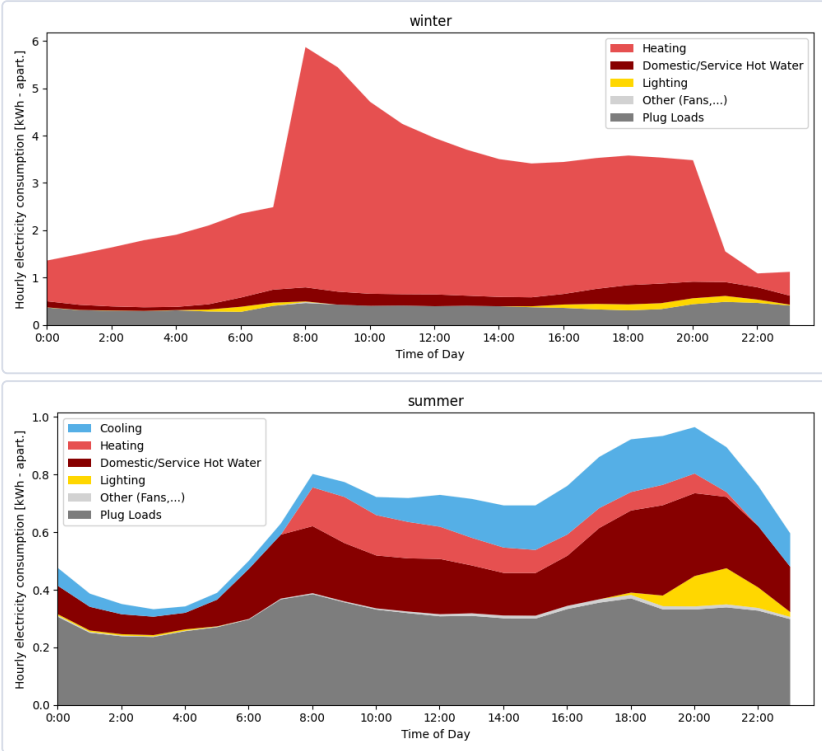
END USE	[kWh/year per apart.]
Heating	9528
Cooling	381
Interior Lighting	237
Exterior Lighting	20
Interior Equipment	4946
Fans	30
Total End Uses	15142

MONTHLY CONSUMPTION BY END USE



Monthly energy consumption breakdown by end use, showing the contribution of heating, cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout the year.

AVERAGE DAILY PROFILE BY USAGE



Average daily consumption profile for winter and summer, showing the contribution of cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout an average winter and summer day.

ENERGY USE INTENSITY

43

kWh/m²/yr

TOTAL CONSUMPTION

15142

kWh/yr

ELECTRICITY

15142

kWh/yr

NATURAL GAS

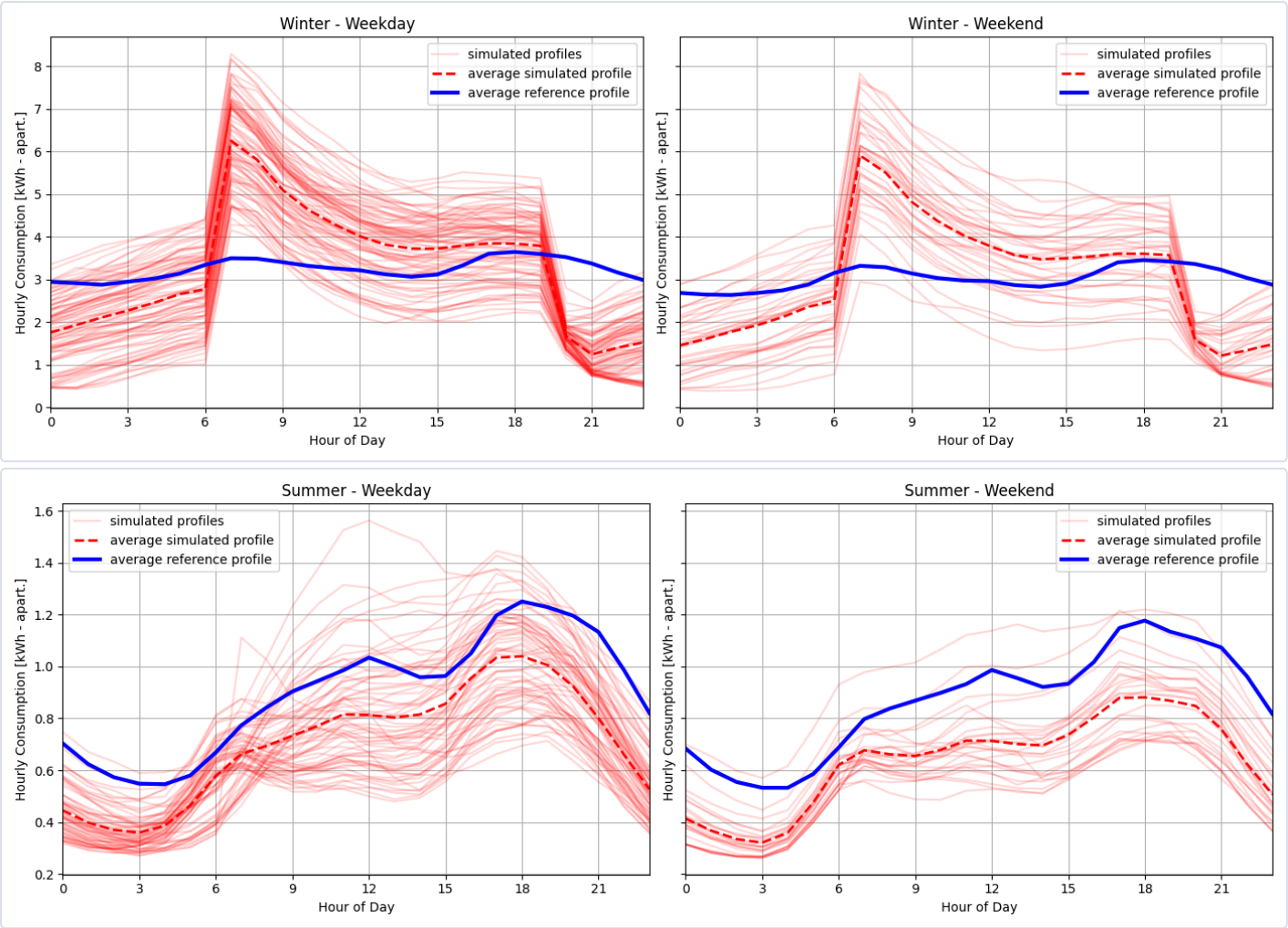
0

kWh/yr

DONNÉES DE RÉFÉRENCE

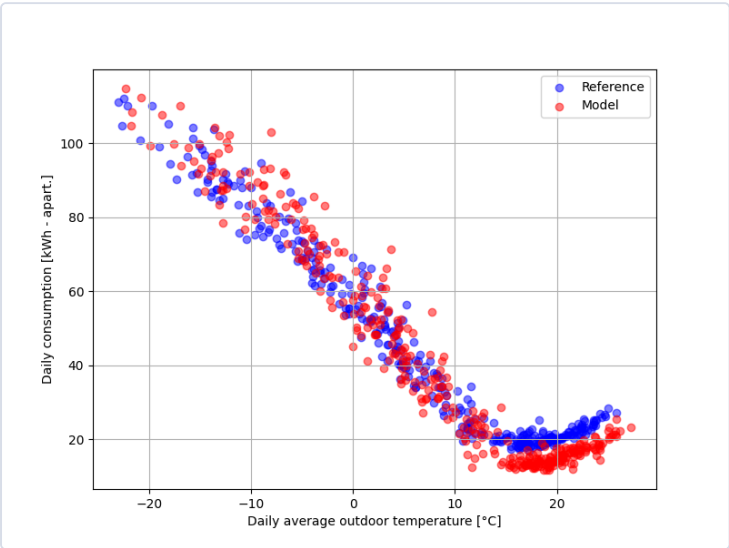
The reference data used for the comparison come from a dataset of more than 60,000 residential electricity meters, combined with metadata. Filters were applied to ensure consistency with the building type and subtype of this card, as well as to exclude end uses such as pool, spa, and electric vehicle charging.

DAILY PROFILE



Daily consumption profiles for summer and winter comparing simulated hourly energy use with the reference data. Red lines represent individual days from the model for the given period, the red dashed line indicates the average simulated profile, and the blue line shows the average profile of the reference data. The figures on the left represent weekdays, while the figures on the right represents weekends.

DAILY CONSUMPTION VS. DAILY AVERAGE OUTDOOR TEMPERATURE



Comparison between modeled daily consumption and reference consumption as a function of outdoor temperature, for all days of the reference year.

COMPARISON TABLE — KEY INDICATORS

INDICATOR	MODEL	REFERENCE	DELTA
Yearly electricity consumption [kWh]	15142	15142	15142
Baseload consumption [kWh/day]	13.9	19.7	-5.8
Electric heating slope [W/K]	-116.2	-107.9	-8.4
Cooling slope [W/K]	46.9	57.4	-10.5
Winter peak demand - AM [kW]	8.3	5.0	3.3
Winter time of peak - AM	7.0	8.0	-1.0
Winter peak demand - PM [kW]	5.8	5.2	0.5
Winter time of peak - PM	12.0	18.0	-6.0